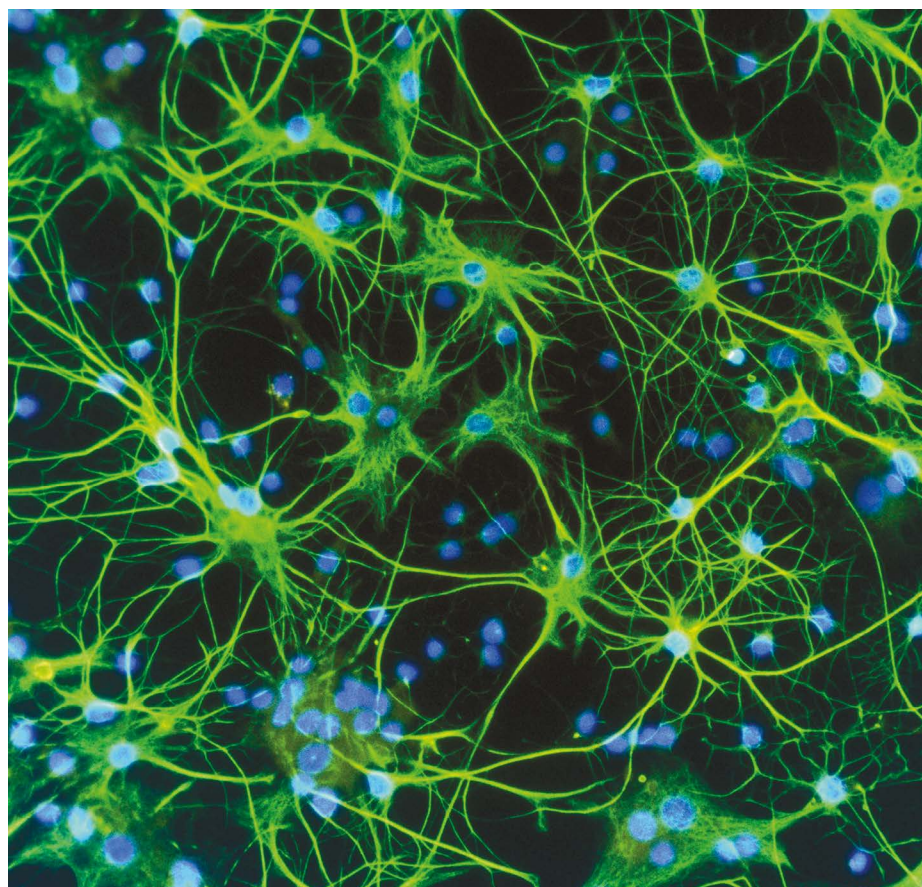




Under a light microscope, brain cells called astrocytes look star-shaped.

THE 'SILENT' BRAIN CELLS THAT SHAPE BEHAVIOUR

Astrocytes make up one-quarter of the brain, but researchers are only now realizing their true value. **By Alison Abbott**

For decades, neuroscientists focused almost exclusively on only half of the cells in the brain. Neurons were the main players, they thought, and everything else was made up of uninteresting support systems.

By the 2010s, memory researcher Inbal Goshen was beginning to

question that assumption. She was inspired by innovative molecular tools that would allow her to investigate the contributions of another, more mysterious group of cells called astrocytes. What she discovered about their role in learning and memory excited her even more.

At the beginning, she felt like an outsider, especially at conferences. She imagined

colleagues thinking, “Oh, that’s the weird one who works on astrocytes,” says Goshen, whose laboratory is at the Hebrew University of Jerusalem. A lot of people were sceptical, she says.

But not any more. A rush of studies from labs in many subfields are revealing just how important these cells are in shaping our behaviour, mood and memory. Long thought of as support cells, astrocytes are emerging as key players in health and disease.

“Neurons and neural circuits are the main computing units of the brain, but it’s now clear just how much astrocytes shape that computation,” says neurobiologist Nicola Allen at the Salk Institute for Biological Studies in La Jolla, California, who has spent her career researching astrocytes and other non-neuronal cells, collectively called glial cells. “Glial meetings are now consistently oversubscribed.”

Out of the shadows

As far back as the nineteenth century, scientists could see with their simple microscopes that mammalian brains included two major types of cell – neurons and glia – in roughly equal numbers.

Twentieth-century technologies generated most of the excitement around neurons. Researchers studying the cells’ electrical activity showed how they create the complex networks that underlie all brain functions.

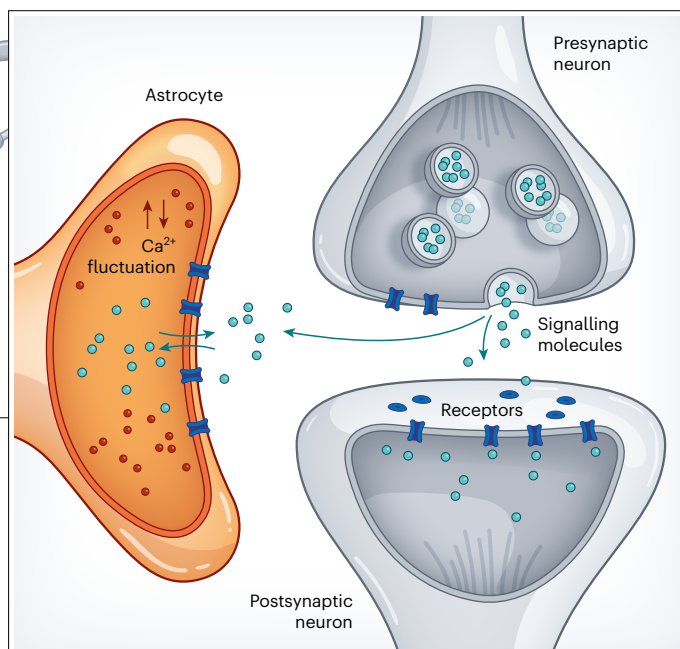
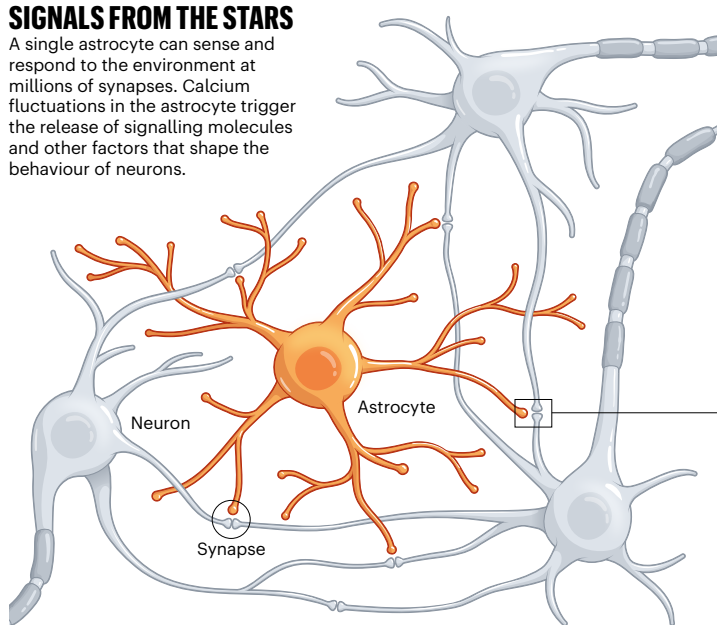
When neurons are activated, electrical signals zap down their length at lightning speed, causing their synapses to release chemical neurotransmitters. Some of these, such as glutamate, excite neighbouring neurons, whereas others, such as GABA (γ-aminobutyric acid), inhibit them. The development of a technique called patch clamping in the 1970s and 1980s, in which electrodes are inserted into individual cells to measure the flow of ions across membranes, let researchers probe this neurotransmission in unprecedented detail.

By contrast, glial cells seemed to be electrically silent and were dismissed as dull by most researchers. Some glia, called oligodendrocytes, coat and insulate neurons. Others, the microglia, provide the brain’s immune system. The many functions of astrocytes proved more elusive.

Although few brain scientists focused on astrocytes until well into the 2000s, some fundamental discoveries were made before that¹. Under the light microscope, astrocytes look star-shaped, stretching between neurons and tiny blood vessels in the brain. They signal blood vessels to increase or decrease their flow according to neuronal activity and extract oxygen and other life-giving molecules from the blood to ship to neurons. They clear up waste products from around the synapses and regulate ion levels there. They extract glutamate from around the neurons to stop the excitatory circuits from going into overdrive, break the neurotransmitter down and then return it

SIGNALS FROM THE STARS

A single astrocyte can sense and respond to the environment at millions of synapses. Calcium fluctuations in the astrocyte trigger the release of signalling molecules and other factors that shape the behaviour of neurons.



building blocks to the neurons.

Astrocytes started to look more interesting when it became possible to measure calcium movements in cells. Those measurements revealed that astrocytes use fluctuations in their calcium levels to signal to each other and to other cells in response to certain molecules in their environment, such as an excess of neurotransmitters (see ‘Signals from the stars’). These calcium signals, which move slowly compared with those that pass between neurons, turn out to have big consequences. They drive many activities, including the release of further signalling molecules, ions, metabolites and other factors that affect how neurons, other glial cells and blood vessels behave.

Because many of the signalling molecules are neurotransmitters, a small group of researchers in the 2010s began to wonder whether astrocytes might contribute to ultrafast electrical transmission in neurons. But over the past decade or so, ever-more-precise research tools have revealed a different, more complex story. Many scientists who had previously focused exclusively on neurons are now using these tools to investigate how astrocytes contribute to animal physiology and behaviour.

No evidence emerged to support a role for astrocytes in ultrafast neurotransmission. Instead, it is becoming ever clearer that astrocytes orchestrate the molecular mix in the environment around synapses on timescales much slower than neuronal signalling, varying that mix according to brain state – how alert or awake the brain is, for example. This, in turn, can determine whether neurons fire in response to a signal coming across the synapse.

Advances in microscopy have shown just how closely astrocytes are associated with neurons. Beyond their star-shaped core, astrocytes have an intricate architecture with

many branches that extend into tiny, delicate structures, called leaflets, that are only tens of nanometres wide. At this resolution, astrocytes, which make up one-quarter of all brain cells, look less like stars and more like bushy spheres, and they fill all the available space between neurons without overlapping with each other². In the human brain, each astrocyte can contact up to two million synapses. What’s more, there are different types of astrocyte in different brain areas³.

This vast reach of astrocytes has profound consequences in the brain, says neuroscientist Baljit Khakh at the University of California, Los Angeles. “Form follows function in biology.” His lab has developed many of the molecular and genetic tools that allow particular calcium signalling pathways in astrocytes to be switched on or off, letting scientists see what each pathway does.

Neurons might transmit the signals that drive brain function, but it is now clear that astrocytes fine-tune those signals by altering the environment around the synapses, says neuroscientist Hongkui Zeng, director of the Allen Institute for Brain Science in Seattle, Washington. “That is why it is important to learn about all brain cells, not just the neurons.”

Clocks, learning and memory

As just one example, astrocytes resolved an open question in the field of circadian biology. How does the body’s master clock, a cluster of cells in the brain called the suprachiasmatic nucleus (SCN) that regulates biological cycles in a roughly 24-hour rhythm, keep its circular loop? “It was a mystery,” says circadian biologist Michael Hastings at the Laboratory of Molecular Biology in Cambridge, UK.

The master-clock system is made up almost exclusively of neurons that release the inhibitory neurotransmitter GABA, and no one

could see how a system that apparently does nothing but suppress neuronal activity could create a daily cycle. Biological clocks need a feedback mechanism so that they can generate rhythmic cycles autonomously, without needing an external signal every day to restart them.

Ten years ago, Marco Brancaccio, a postdoctoral researcher working with Hastings, read about a glutamate detector – a fluorescent probe called ‘glue sniffer’, or iGluSnFR. Brancaccio proposed using the detector to sniff out any glutamate that might be present in the mouse brain slices the team was studying.

“I told Marco it would be a waste of time, because there wouldn’t be any,” recalls Hastings. “Fortunately he ignored me.”

Brancaccio, who is now at Imperial College London, found plenty of glutamate – and to general astonishment, found that its levels were as rhythmic as those of GABA. But whereas the GABA levels were highest during the day, glutamate levels peaked at night⁴.

“We scratched our heads. Where could that glutamate be coming from?” says Hastings. It didn’t take them long to discover through a literature search that the source must be astrocytes, a cell type they hadn’t thought much about before.

This set the researchers on a series of ever-more sophisticated experiments. They ultimately concluded that astrocytes support the SCN clock by switching on their uptake systems for GABA by day and turning them off at night^{4,5}.

Particularly intriguing are studies that are revealing how astrocytes support the many subtleties of learning and memory. In one of her first key experiments with the new tools, Goshen and her team were amazed to discover how astrocytes in the brains of mice encode information about the spatial location of

rewards. When a mouse had already learnt where to find a water reward, the calcium activity in its astrocytes gradually increased as it approached the reward, the scientists found. But the researchers measured no increase when the mouse moved towards the same reward in a new environment⁶. The finding raises interesting questions about how astrocytes are involved in the encoding of spatial memory.

Earlier this year, groups in Japan and the United States reported that astrocytes help with the stabilization and recall of memories primed by fear^{7,8}. Because the astrocytes' signals are much slower than the electrical signals of neurons – sometimes developing over hours or even days rather than milliseconds – they are well suited to bridging the time gap between learning and remembering, says neuroscientist Jun Nagai at the RIKEN Centre for Brain Science in Wako, Japan, who led one of the studies. “Think of them as the brain’s long-exposure camera: they capture the trace of meaningful events that might otherwise fade too fast.”

Because neurons and astrocytes work together to process information, researchers have started to wonder whether astrocytes drive or exacerbate diseases that have been generally considered to be neuronal.

Khakh has little doubt they do. Using a combination of techniques, his team has shown⁹ in mice that activating or blocking particular calcium signals in astrocytes can promote or suppress behaviours that are characteristic of some psychiatric conditions, including obsessive–compulsive disorder. Astrocytes seem to modify the neuronal circuits that drive these behaviours and so could be targets for therapies, says Khakh.

Astrocytes might also have potential as targets for therapies for neurological diseases, even if they might not deliver full cures. Khakh's team has focused on Huntington's disease, which is caused by a mutation in a single gene. It is a devastating neurodegenerative illness characterized by psychiatric, cognitive and motor symptoms.

Using a mouse model of the disease and analysing human data, the team identified a set of genes in astrocytes in the brain's striatum that show reduced expression in Huntington's disease¹⁰. These genes were involved mostly in maintaining the extracellular environment that supports neurons. When the researchers used molecular tools to boost the expression of some of these genes in mice, certain behaviours were corrected. The findings could be clinically relevant for treating early-stage psychiatric and cognitive symptoms of Huntington's, including difficulty with attention and planning, apathy and depression, says Khakh.

“Neurons clearly drive the disease, but drug developers should not just target neurons

because they, by definition, exist in tissue which is itself dysfunctional,” he says.

The same could be said for Alzheimer's disease. This common neurodegenerative condition, characterized by the presence of amyloid plaques in the brain, has long been known to involve microglia, which release inflammatory molecules in an attempt to clear the plaques. The resulting inflammation becomes ever more futile as the disease advances, and the inflammation itself begins to add to tissue destruction.

Neuroscientist Bart De Strooper, at University College London, and his team have now revealed in a mouse model of Alzheimer's disease that astrocytes and microglia operate together almost as a single unit to accelerate this destruction¹¹. The team has also shown that astrocytes might be involved at the beginning of the disease, long before plaques get laid down. At this kick-off stage, neurons become pathologically hyperactive, and



THINK OF THEM AS THE BRAIN'S LONG-EXPOSURE CAMERA.”

calcium signals in astrocytes fall at the same time. When the team used a molecular tool to correct this drop in calcium, neuronal activity normalized and the mice no longer displayed some early disease symptoms, such as sleep disturbances¹².

In a preprint study¹³ using another animal model of Alzheimer's, Goshen's team activated a calcium-signalling pathway in astrocytes. This fully restored normal memory in mice that were showing signs of neurodegeneration.

Scientists are only beginning to understand the details of how glial cells and neurons interact in Alzheimer's. “But I think it is already clear that future therapies will need to target all of the cells that are involved, not just neurons,” says De Strooper.

Better brain models

The excitement over astrocytes is slowly starting to draw in computational neuroscientists who use experimental data to test theories about how the brain works.

Most modelling of astrocytes has been small scale – up to 100 or so astrocytes alongside neurons, says computational neuroscientist Marja-Leena Linne at Tampere University in Finland. She and her team, along with scientists from the Jülich Research

Centre in Germany, have built what is so far the most extensive astrocyte simulator, incorporating molecular, cellular, morphological and connectivity data from diverse sources. The scientists have integrated this tool into a digital platform that includes other tools for simulating neural networks. (The platform is supported by EBRAINS, a European infrastructure for neuroscience research at the Jülich Supercomputing Centre.) Neuroscientists can now use any connectivity data to model up to one million astrocytes and neurons¹⁴.

Some researchers who have been working on astrocytes for decades speculate about a possible evolutionary role for astrocytes in enabling the special cognitive capabilities of humans. “Much of the rush of recent data is consistent with this speculation,” says astrocyte biologist Alexej Verkhratsky at the University of Manchester, UK. Human astrocytes are structurally more complex than are those in monkeys and rodents; astrocyte-like cells in flies are even simpler. A human astrocyte covers a volume that is at least an order of magnitude larger than that of its rat counterpart and has ten times more branches.

That seems to translate into intelligence, says Verkhratsky. Neurobiologist Maiken Nedergaard at the University of Rochester Medical Center in New York and her colleagues engrafted human astrocytes into mice, and found not only that the integrated human astrocytes retained their large size and complexity, but also that the chimeric mice were markedly smarter than control mice in various memory tasks¹⁵.

After decades in the shadows, astrocytes are opening many avenues of research. “The amount of literature now coming out from different labs working in different areas has led most sceptics to appreciate that astrocytes are doing something that they should think about,” says Allen.

Alison Abbott is a writer in Munich, Germany.

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